NAAN MUDHALVAN PROJECT

TITLE NAME: FARM MANAGEMENT SYSTEM (FMS)

PROJECT REPORT

DEVELOPED BY

M. NALINI

V. RAMYA

M.A. SUMAIYA

V. SUVALAKSHMI

BCA DEPARTMENT III-YEAR

KRISHNASAMY COLLEGE OF SCIENCE, ARTS AND MANAGEMENT FOR WOMEN

USER STORY

In the Farm Management System (FMS), farmers, farm managers, and administrators use a centralized Salesforce platform to manage farm, crop, farmer, and buyer information. The system allows users to record and update agricultural data easily with proper validation rules to avoid incorrect entries.

It also provides reports and dashboards to analyze crop production, farm performance, and sales activities. The platform ensures secure data storage with role-based access so that sensitive information is protected. This system reduces manual paperwork, improves accuracy, and supports better farm management decisions.

1. INTRODUCTION

1.1 Project Overview

The **Farm Management System (FMS)** is a Salesforce-based application developed to manage agricultural data digitally. It helps store and organize information related to farms, crops, farmers, and buyers.

Using Salesforce features such as **custom objects, relationships, reports, and dashboards**, the system helps users monitor farm activities and analyze performance in real time.

1.2 Purpose

The main purpose of this project is to:

• Digitize farm data management
• Improve data accuracy
• Reduce manual record keeping
• Generate real-time reports and dashboards
• Help farmers and managers make better decisions

2. IDEATION PHASE

2.1 Problem Statement

User	Problem	Reason	Result
Farmer	Managing farm records	Records stored manually	Data errors
Farm Manager	Monitoring production	Data scattered in files	Delay in reports
Buyer	Access crop information	No centralized system	Uncertain information
Admin	Managing system security	Manual access control	Security risks

2.2 Empathy Map

Target Users

• Farmers
• Farm Managers
• Buyers
• System Administrators

Analysis

Think & Feel – Need accurate farm data
Hear – Advice from agricultural experts
See – Paper records and spreadsheets
Say & Do – Manually update records
Pain – Data loss and errors
Gain – Automated system with fast reporting

2.3 Brainstorming

Possible solutions discussed:

• Manual farm records
• Excel-based system
• Web application
• Mobile application
• Salesforce-based management system

Selected Idea

Salesforce-based **Farm Management System**

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

1 Login to system
2 Open Farm Management App
3 Enter farm and crop data
4 Register farmers and buyers
5 Generate reports
6 Analyze dashboard results
7 Make decisions based on data

3.2 Solution Requirements

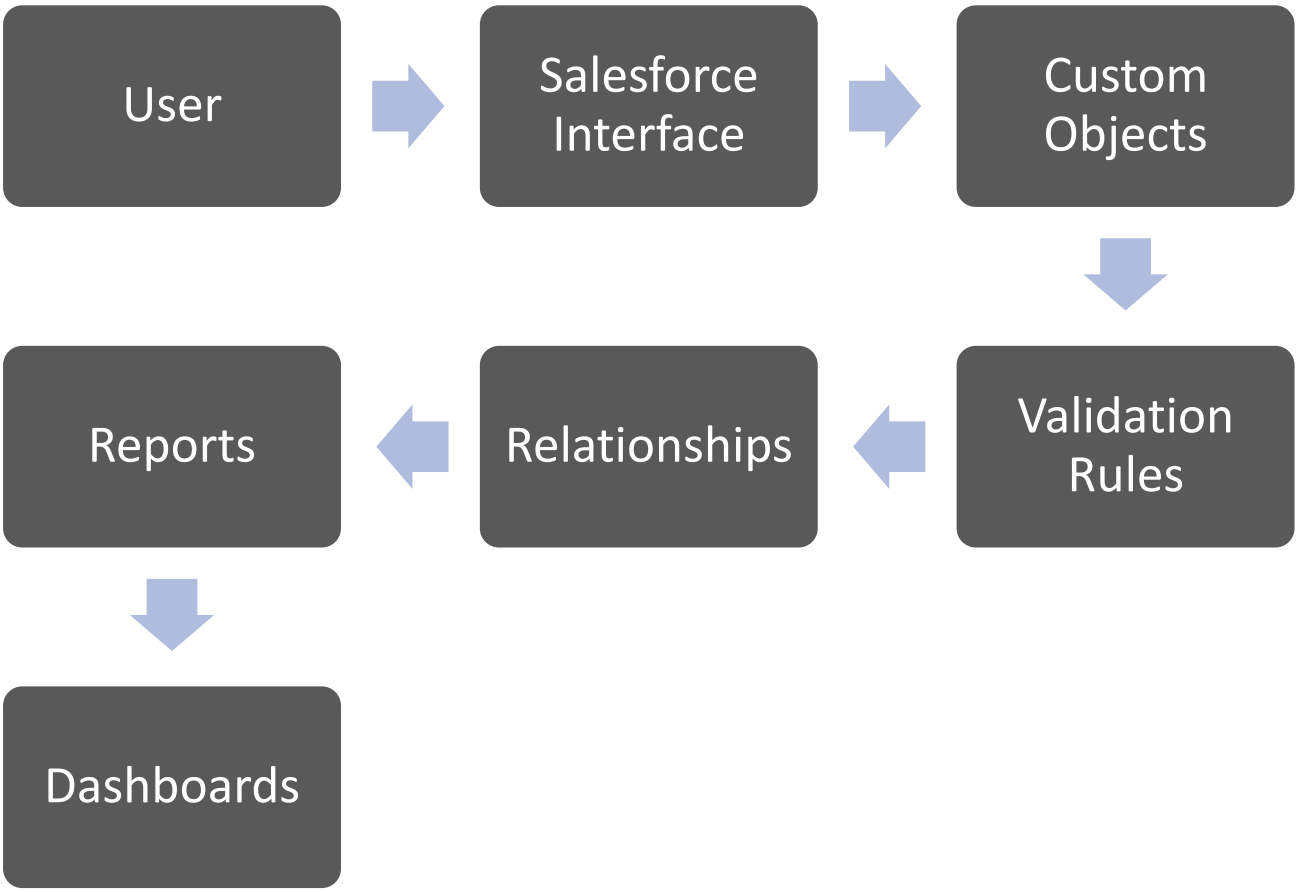
Functional Requirements

FR1 User Login and Authentication
FR2 Manage Farms, Crops, Farmers, Buyers
FR3 Data entry with validation
FR4 Relationship between objects
FR5 Generate Reports
FR6 Dashboard Visualization
FR7 User Role Management

Non Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

3.3 Data Flow Diagram



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports and Dashboards

4. PROJECT DESIGN

4.1 Problem Solution Fit

Traditional farming records are usually maintained in paper registers, which can lead to data loss and errors.

The **Farm Management System** solves this problem by storing all data in Salesforce cloud storage. Reports and dashboards provide quick insights into farm productivity and sales.

4.2 Proposed Solution

Problem: Manual farm record management.

Solution:
A Salesforce-based system that stores farm, crop, farmer, and buyer data in one centralized platform.

Benefits:

• Accurate data storage
• Automated validations
• Faster report generation
• Better decision making

4.3 Solution Architecture

System Architecture Layers

1 Presentation Layer – Salesforce Lightning UI
2 Application Layer – Validation Rules and Automation
3 Data Layer – Custom Objects
4 Analytics Layer – Reports and Dashboards

5. PROJECT PLANNING

5.1 Project Steps

• Create Salesforce Developer Account
• Create Custom Objects
• Create Fields and Relationships
• Configure Tabs and Lightning App
• Create Profiles and Roles
• Create Users
• Configure Sharing Rules
• Create Reports and Dashboards
• Test the application

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As a system admin, I want to set up a Salesforce developer account so that the Farm Management System can be developed and deployed.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create objects like Farm, Crop, Farmer, and Buyer to store farm-related data properly.	5	High	Member2
Sprint-2	Application Setup	USN-3	As a staff member, I want a Lightning App with tabs for farm modules so that I can easily access the system.	5	High	Member3
Sprint-3	Validation	USN-4	As a system, I want to validate farm and crop details to avoid incorrect or incomplete data.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications to reduce manual work.	3	High	Member4
Sprint-4	Security	USN-6	As an admin, I want role-based access for farmers, managers, and buyers to keep data secure.	5	High	Member2

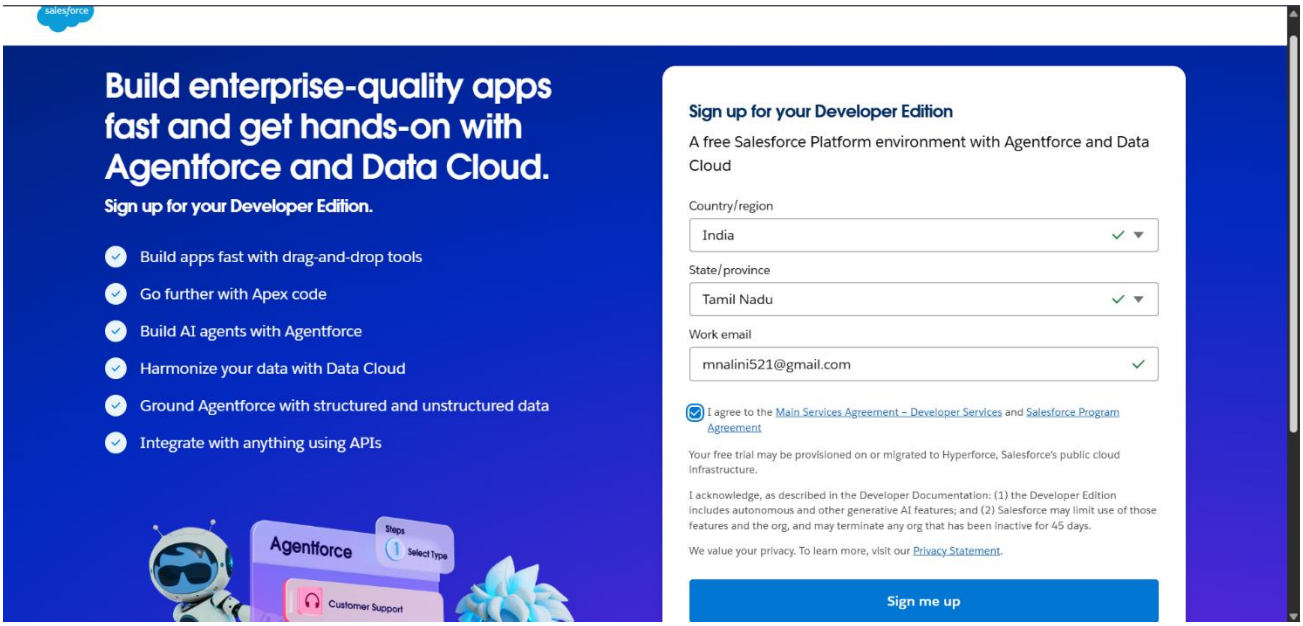
Sprint-5	Reports	USN-7	As a farm manager, I want reports for crop yield, sales, and buyer transactions to monitor farm performance.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing productivity and sales trends for better decision making.	4	Medium	Member4

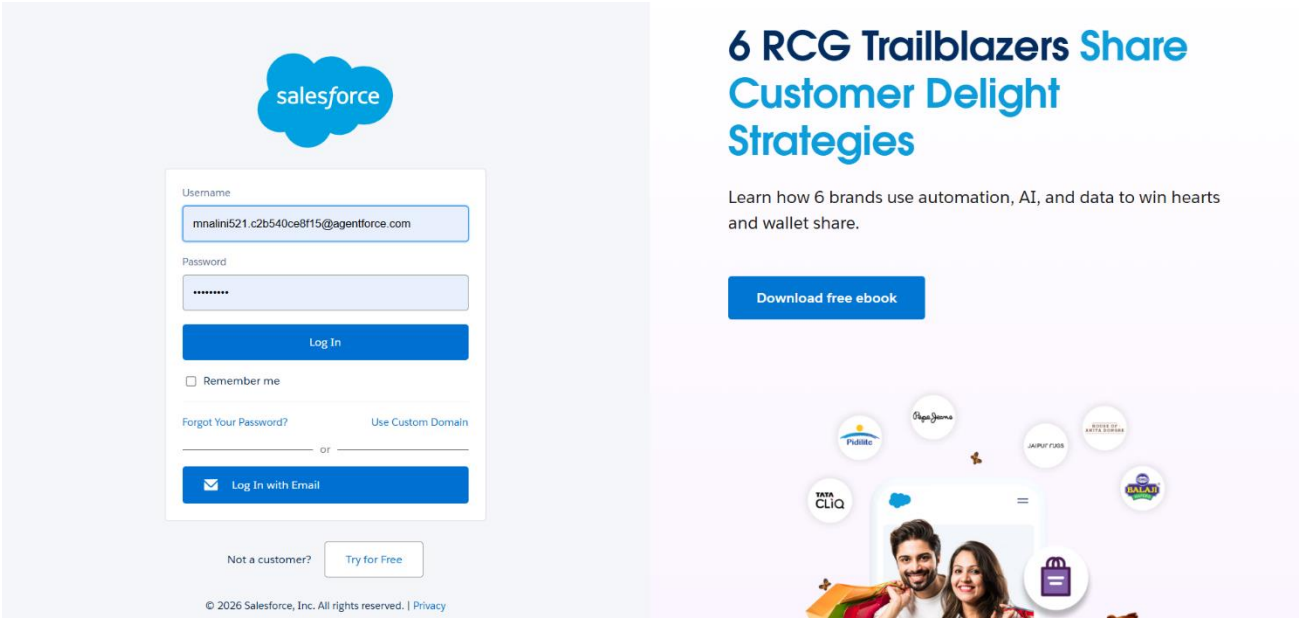
6. PROJECT DEVELOPMENT

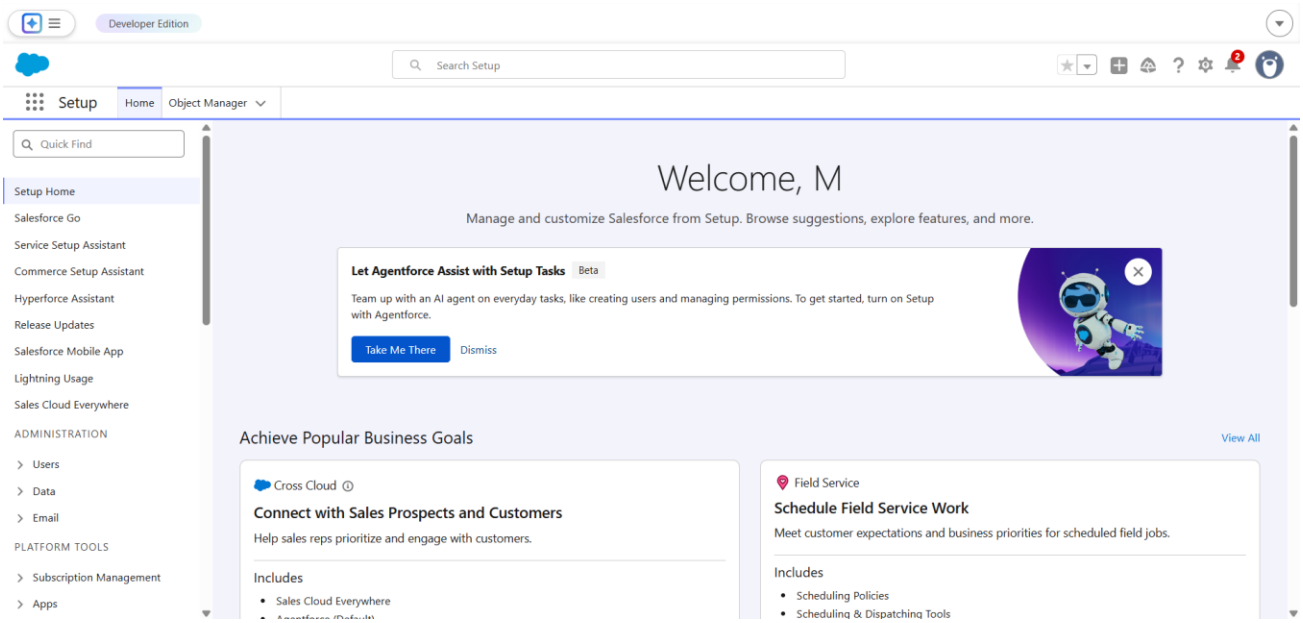
What is Salesforce

Salesforce is a cloud-based CRM platform that helps organizations manage data, automate processes, and analyze business performance.

It allows users to create applications using objects, fields, reports, and dashboards without complex coding.



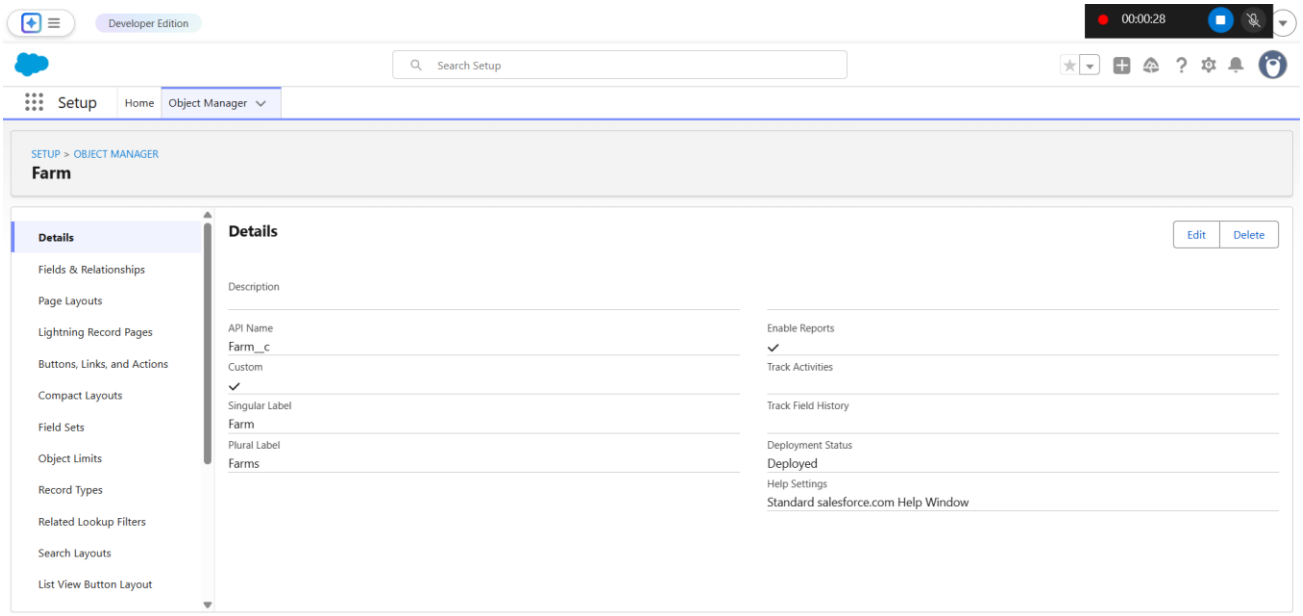




Custom Objects Used

In this project four custom objects were created:

- Farm
- Crop
- Farmer
- Buyer


Developer Edition
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Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Farm
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Details
Edit
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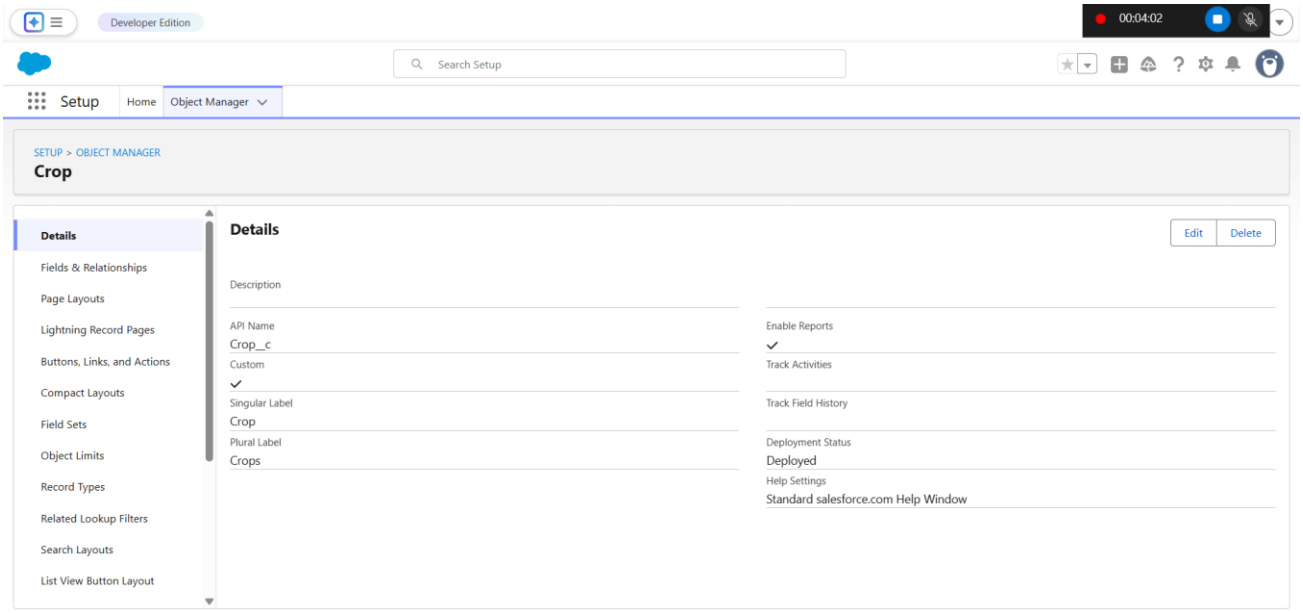
Description
API Name
Farm__c
Custom
Singular Label
Farm
Plural Label
Farms
Enable Reports
Track Activities
Track Field History
Deployment Status
Deployed
Help Settings
Standard salesforce.com Help Window


Developer Edition
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Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Crop
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Details
Edit
Delete
Description
API Name
Crop__c
Custom
Singular Label
Crop
Plural Label
Crops
Enable Reports
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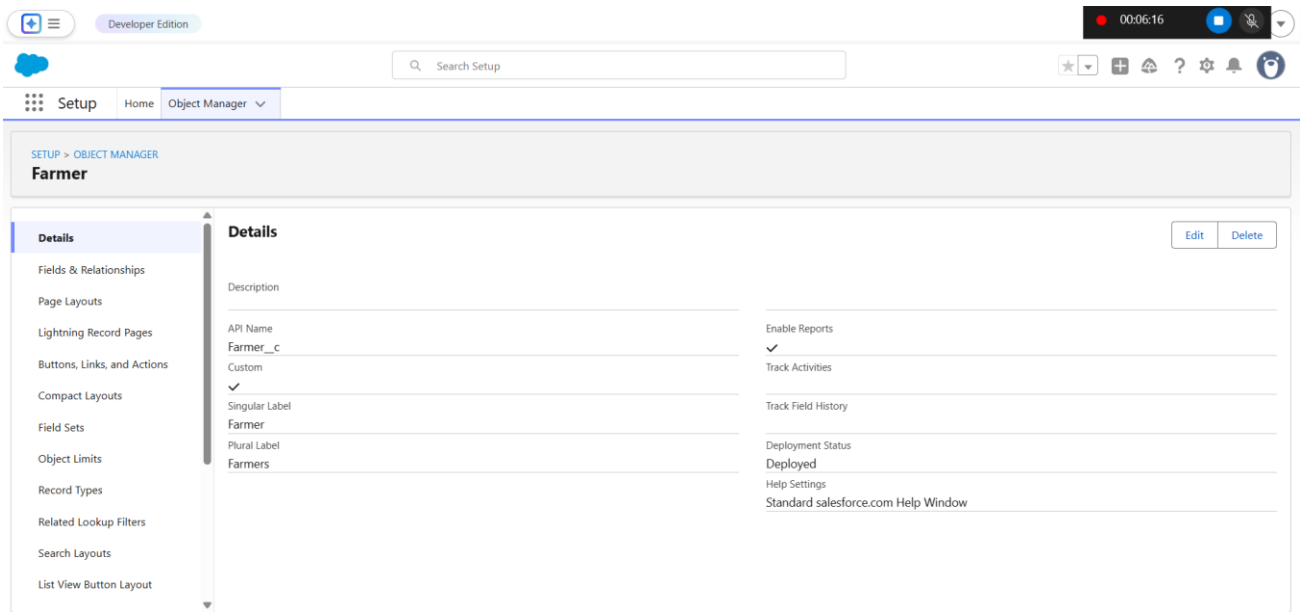
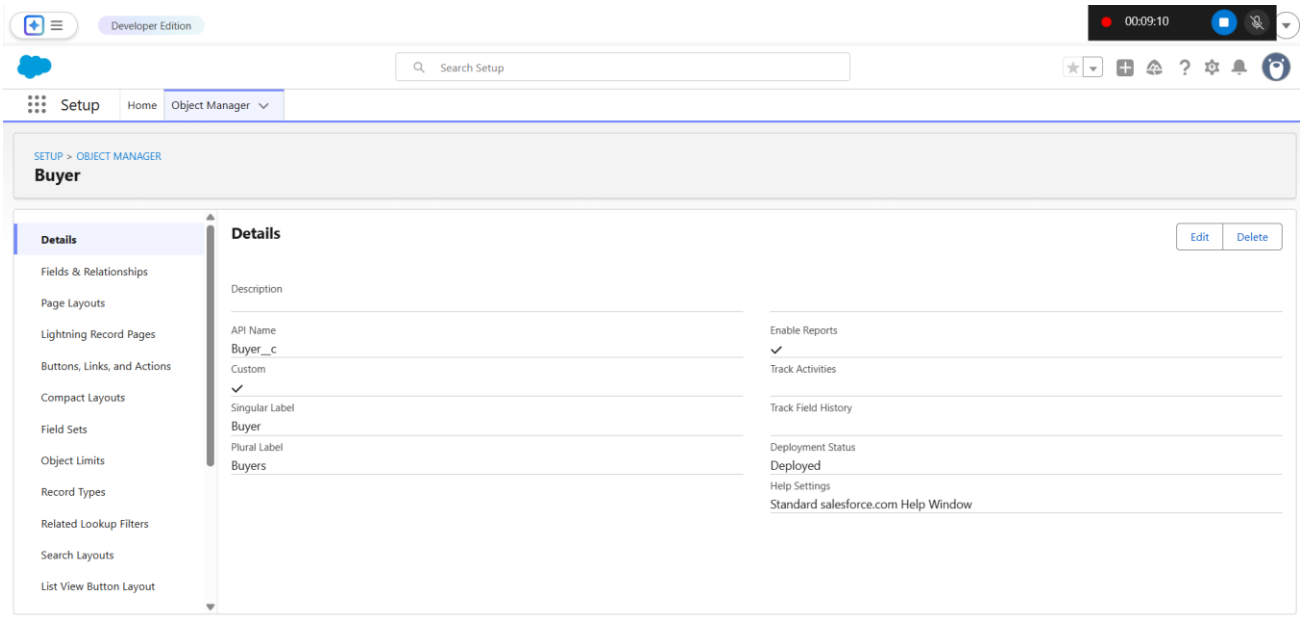
Help Settings
Standard salesforce.com Help Window





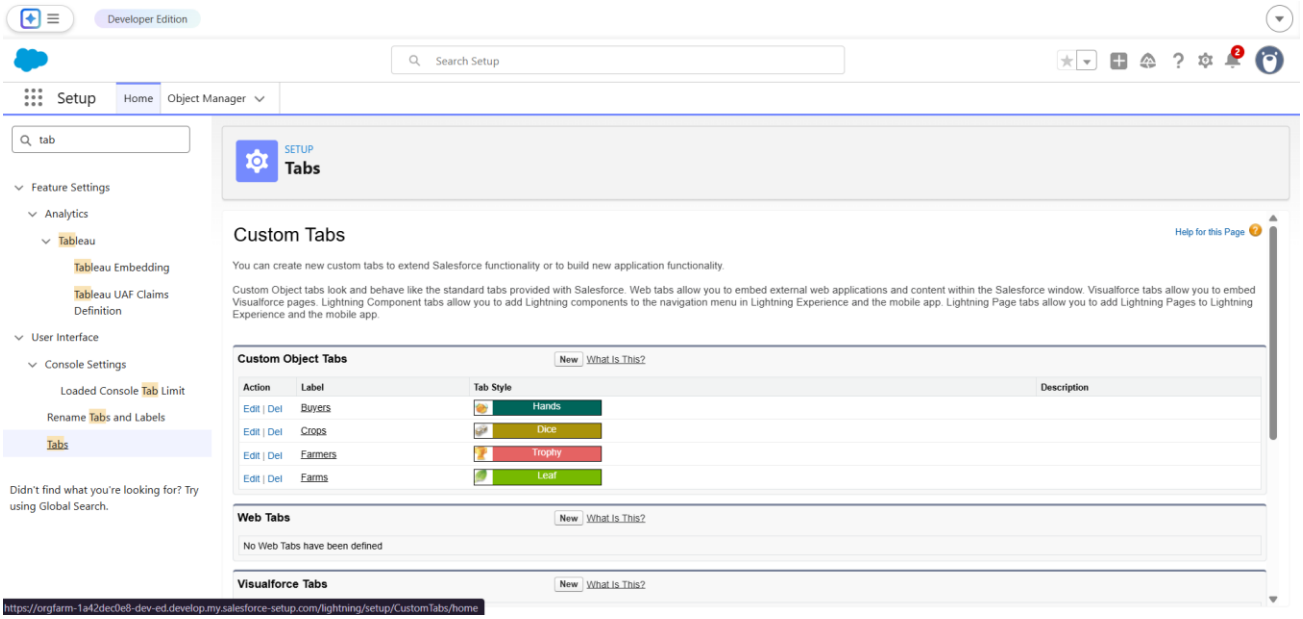
(Screenshot – Object Creation)

Tabs

Tabs were created for each object so users can easily access the data.

Tabs created:

- Farm
- Crop
- Farmer
- Buyer



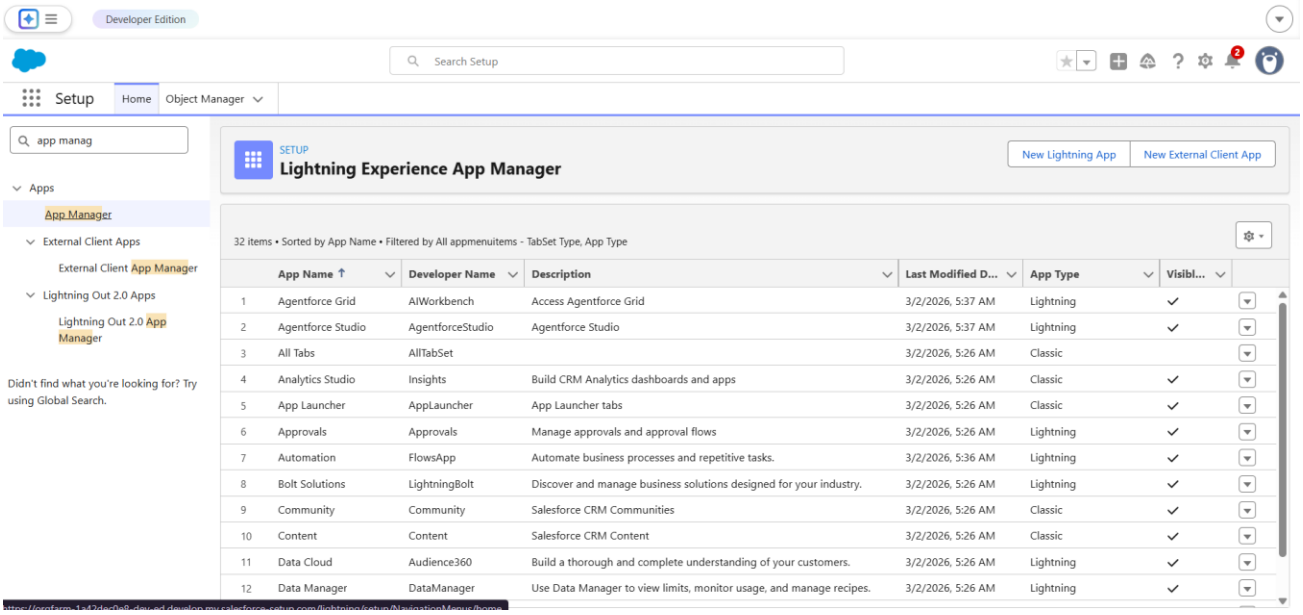
(Screenshot – Tabs)

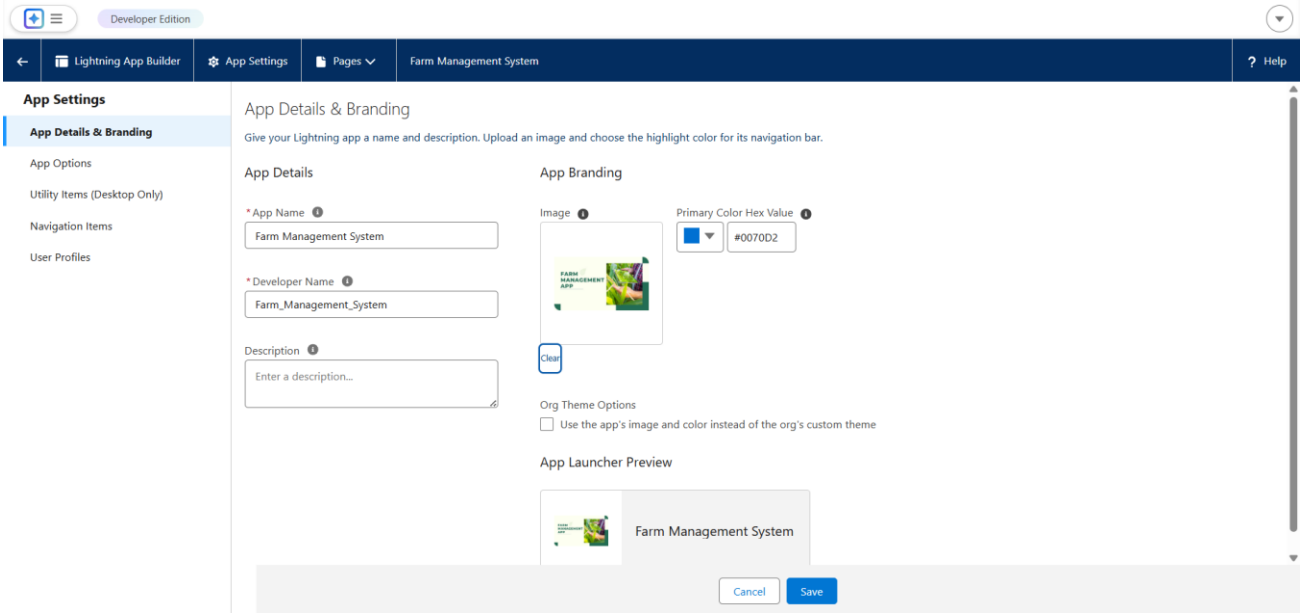
LIGHTNING APP

A **Farm Management System Lightning App** was created using App Manager.

The following tabs were included:

- Farm
- Crop
- Farmer
- Buyer
- Reports
- Dashboards





(Screenshot – Lightning App)

FIELDS AND RELATIONSHIPS

Fields were created for each object to store information.

Examples:

Farm Object
• Farm Type
• Soil Quality
• Irrigation Type
• Location
• Size

Crop Object
• Crop Name
• Planting Date
• Harvest Date
• Yield

Relationships were created between objects such as **Farmer and Farm**, and **Farm and Crop**.

(Screenshot – Fields and Relationships)

Developer Edition

00:21:20

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Farm

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

Fields & Relationships

5 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Type	Farm_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

SETUP > OBJECT MANAGER

Crop

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

Fields & Relationships

12 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Actual Yield	Actual_Yield__c	Currency(18, 0)		
Created By	CreatedById	Lookup(User)		
Crop Name	Crop_Name__c	Text(80)		
Crop Name	Name	Text(80)		✓
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)		
Farm	Farm__c	Lookup(Farm)		✓
Harvest Date	Harvest_Date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		

Developer Edition

00:11:39

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Farmer

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
District	District__c	Picklist	State	
Farmer Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
State	State__c	Picklist		
Village	Village__c	Picklist		

Developer Edition

00:12:07

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Farmer

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

Farmer Field Dependencies

Help for this Page

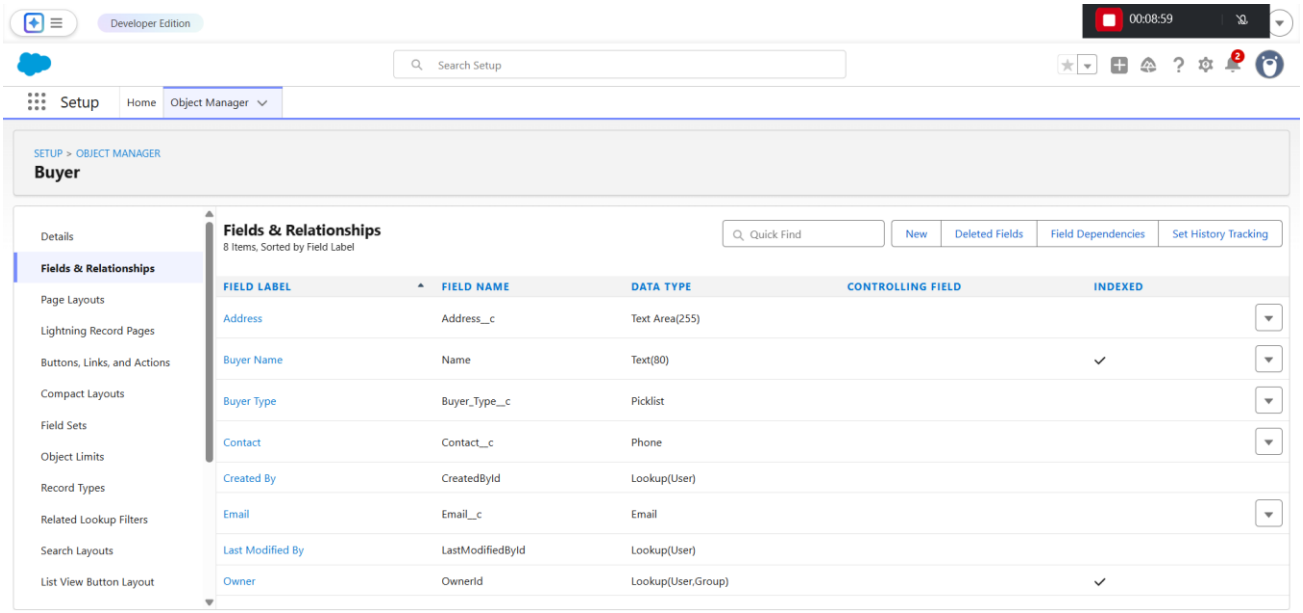
« Back to Custom Object: Farmer

This page allows you to define dependencies between fields (e.g., dependent picklists).

Field Dependencies

New

Action	Controlling Field	Dependent Field	Modified By
Edit | Del	State	District	M Nalini, 3/5/2026, 1:55 AM



PROFILES AND ROLES

Profiles and roles were created to control user access.

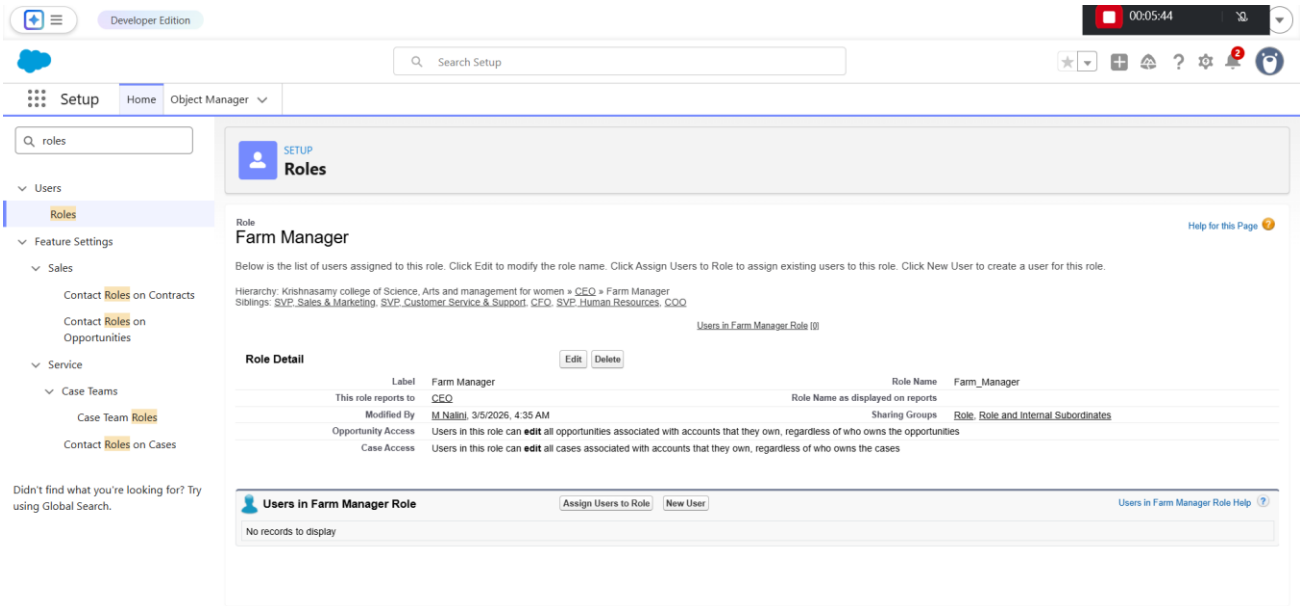
Profiles Created:

- Farm Manager
- Sales Department

Role Created:

- Farm Manager

(Screenshot – Profiles and Roles)



Developer Edition

Search Setup

Setup | Home | Object Manager

profiles

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP
Profiles

Profile
Farm Manager

Help for this Page

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

Login IP Ranges [0] | Enabled Apex Class Access [0] | Enabled Visualforce Page Access [0] | Enabled External Data Source Access [0] | Enabled Named Credential Access [0] | Enabled External Credential Principal Access [0] | Enabled Custom Metadata Type Access [0] | Enabled Custom Setting Definitions Access [0] | Enabled Flow Access [0] | Enabled Service Presence Status Access [0] | Enabled Custom Permissions [0]

Profile Detail Edit | Clone | Delete | View Users

Name	Farm Manager		
User License	Salesforce	Custom Profile	✓
Description			
Created By	M Nalini, 3/5/2026, 4:34 AM	Modified By	M Nalini, 3/5/2026, 8:31 AM

Page Layouts

Standard Object Layouts

Global	Global Layout [View Assignment]	Location Group Assignment	Location Group Assignment Layout [View Assignment]
Email Application	Not Assigned [View Assignment]	Macro	Macro Layout [View Assignment]
Home Page Layout	Home Page Default [View Assignment]	Object Milestone	Object Milestone Layout [View Assignment]
Account	Account Layout [View Assignment]	Operating Hours	Operating Hours Layout [View Assignment]

Developer Edition

Search Setup

Setup | Home | Object Manager

profiles

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP
Profiles

Profile
Sales Department

Help for this Page

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

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Profile Detail Edit | Clone | Delete | View Users

Name	Sales Department		
User License	Salesforce	Custom Profile	✓
Description			
Created By	M Nalini, 3/5/2026, 4:29 AM	Modified By	M Nalini, 3/5/2026, 8:31 AM

Page Layouts

Standard Object Layouts

Global	Global Layout [View Assignment]	Location Group Assignment	Location Group Assignment Layout [View Assignment]
Email Application	Not Assigned [View Assignment]	Macro	Macro Layout [View Assignment]
Home Page Layout	Home Page Default [View Assignment]	Object Milestone	Object Milestone Layout [View Assignment]
Account	Account Layout [View Assignment]	Operating Hours	Operating Hours Layout [View Assignment]

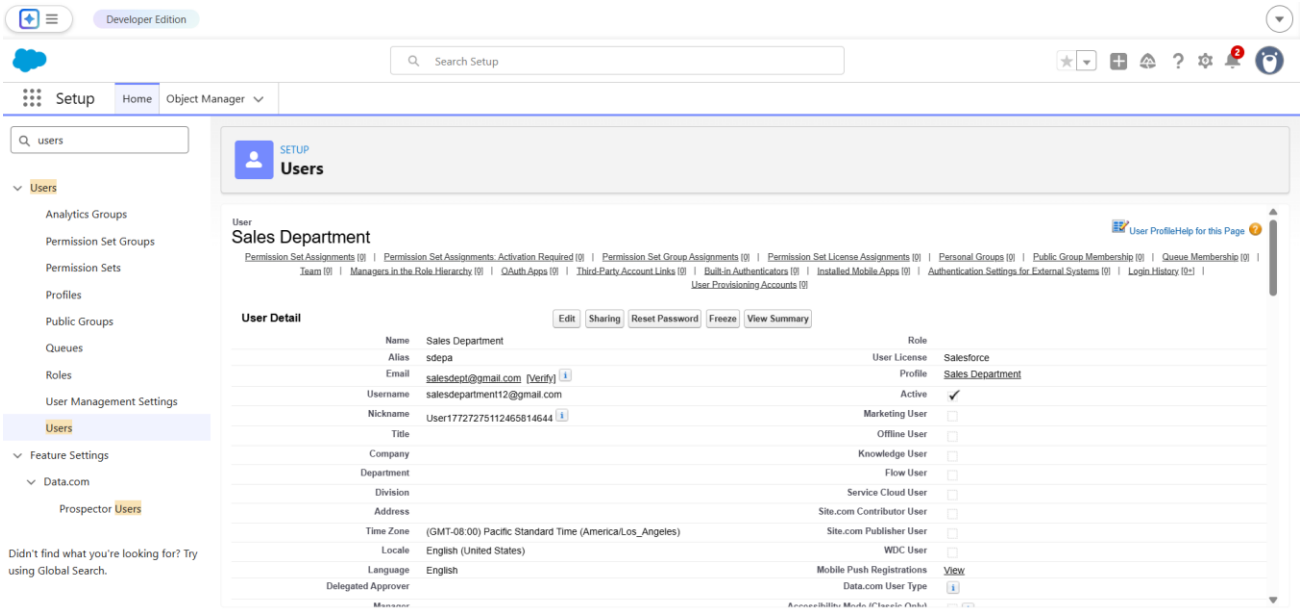
USERS

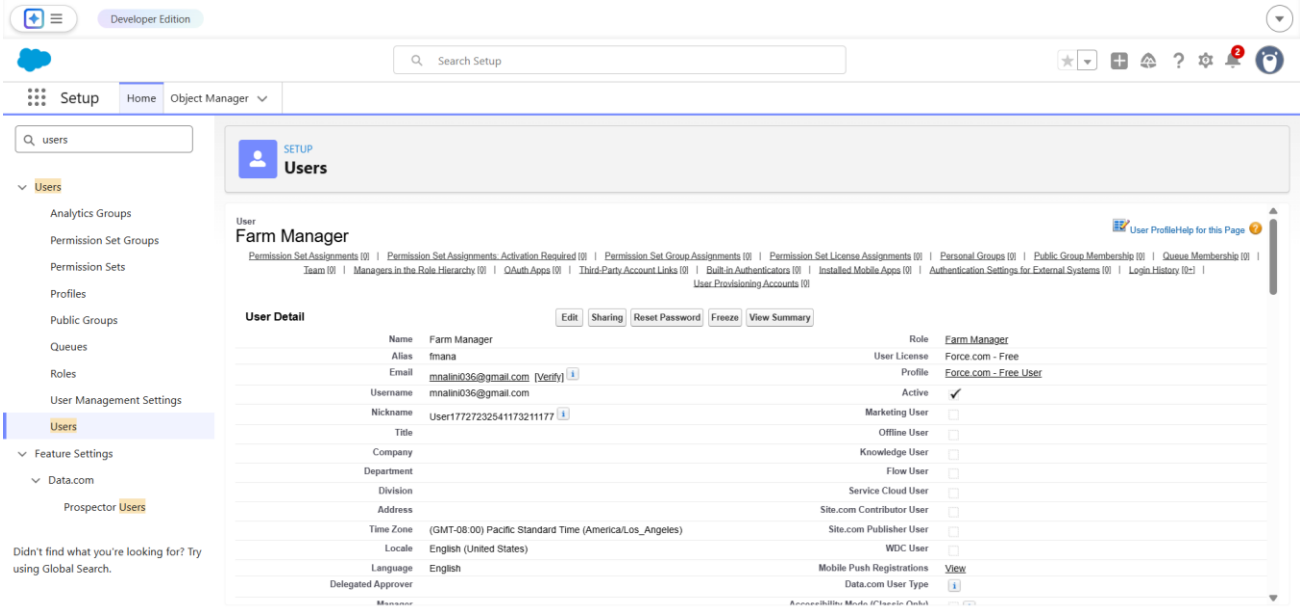
Users were created to access the system.

Example:

Farm Manager User

(Screenshot – User Creation)





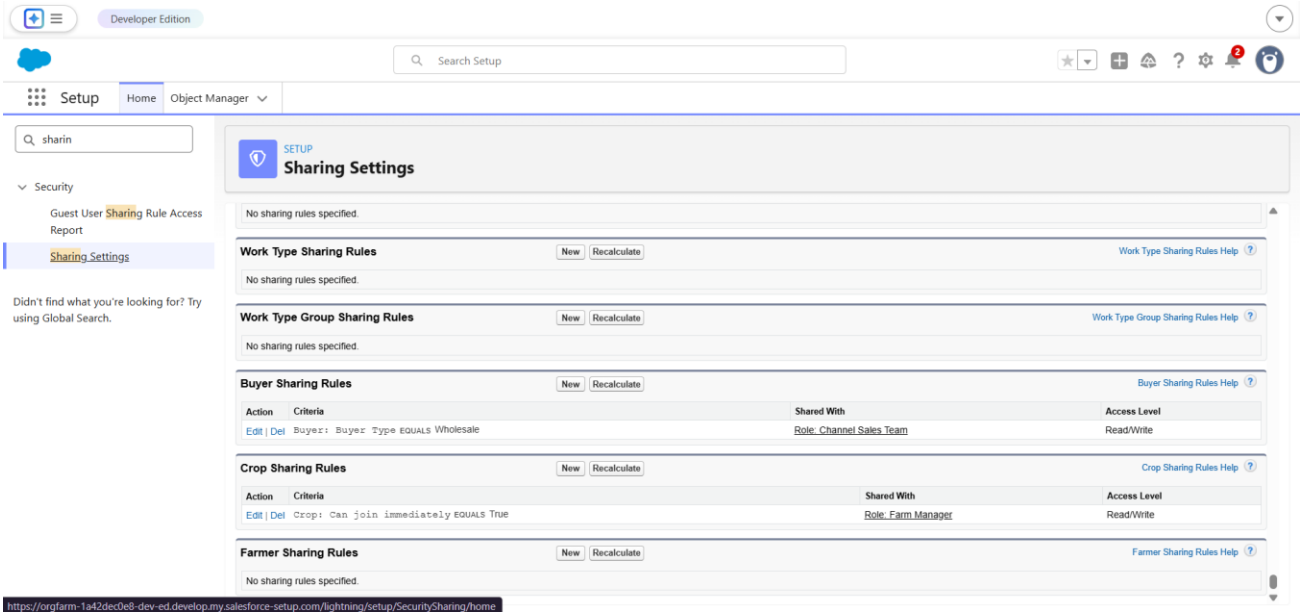
SHARING RULES

Sharing rules were created to allow data access between users.

Example:

Crop records shared with Farm Manager.

(Screenshot – Sharing Rules)



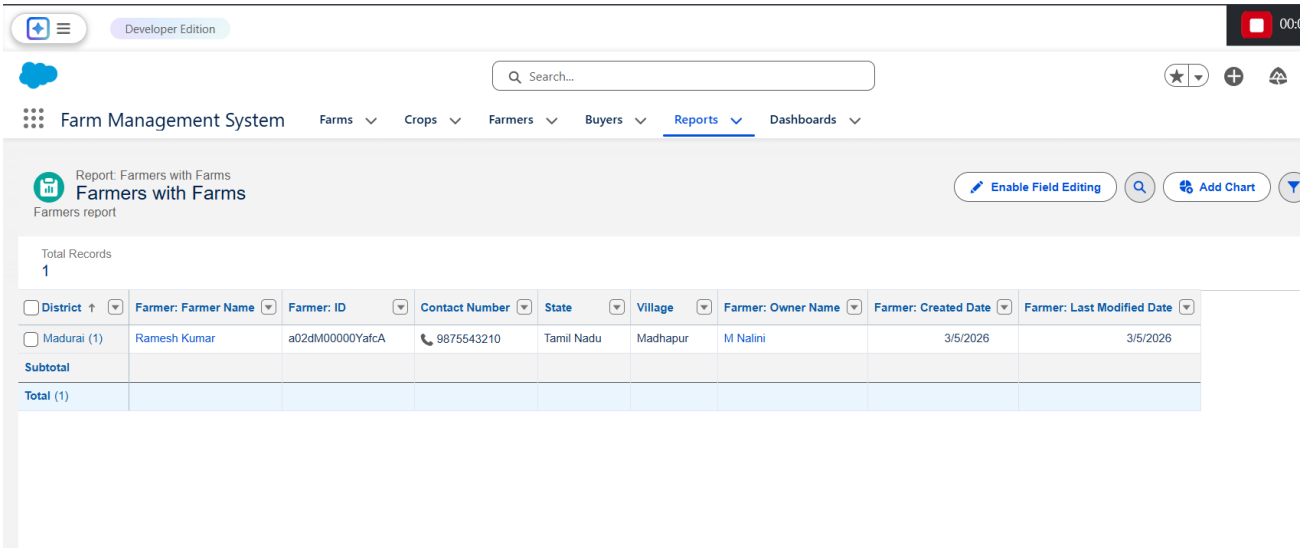
REPORTS

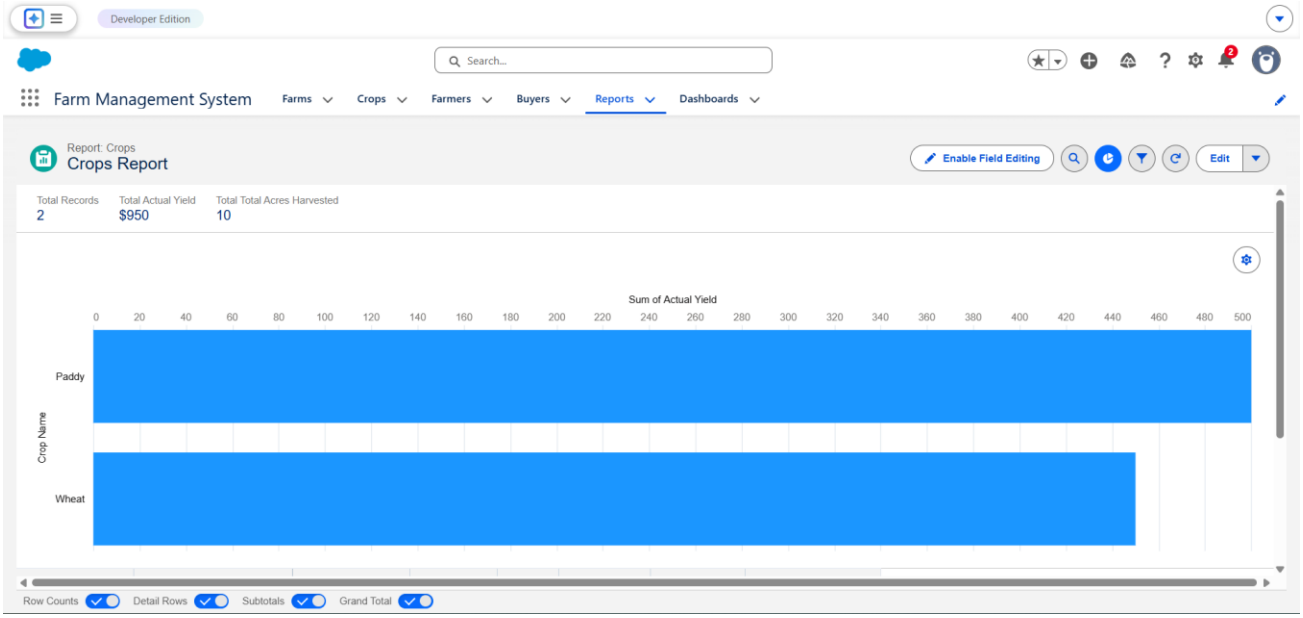
Reports help analyze stored data.

Example Reports:

- Farmers with Farms
- Crops with Buyers

(Screenshot – Reports)


Developer Edition
Search...
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Report: Farmers with Farms
Farmers with Farms
Farmers report
Enable Field Editing
Add Chart
Total Records
1
District
Farmer: Farmer Name
Farmer: ID
Contact Number
State
Village
Farmer: Owner Name
Farmer: Created Date
Farmer: Last Modified Date
Madurai (1)
Ramesh Kumar
a02dM00000YafcA
9875543210
Tamil Nadu
Madhapur
M Nalini
3/5/2026
3/5/2026
Subtotal
Total (1)


Developer Edition
Search...
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Report: Crops
Crops Report
Enable Field Editing
Edit
Total Records
2
Total Actual Yield
$950
Total Total Acres Harvested
10
Sum of Actual Yield
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
320
340
360
380
400
420
440
460
480
500
Crop Name
Paddy
Wheat
Row Counts
Detail Rows
Subtotals
Grand Total

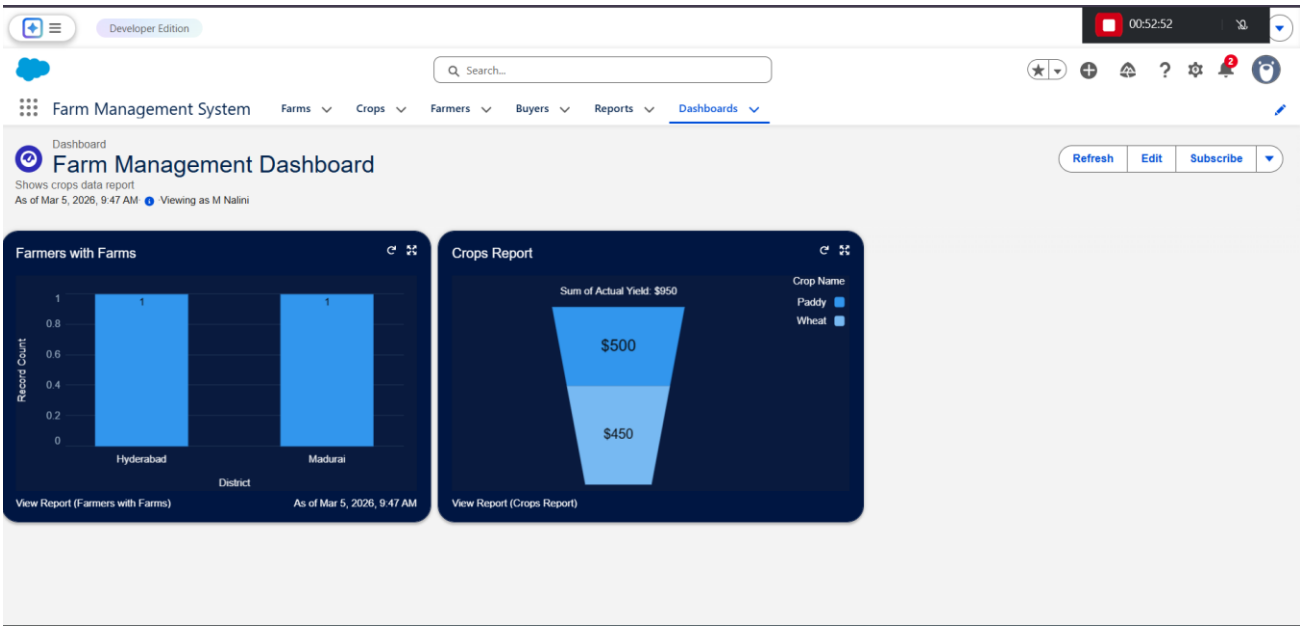
DASHBOARDS

Dashboards display summarized data using charts and graphs.

Example Dashboard:

• Farmers with Farms

(Screenshot – Dashboard)

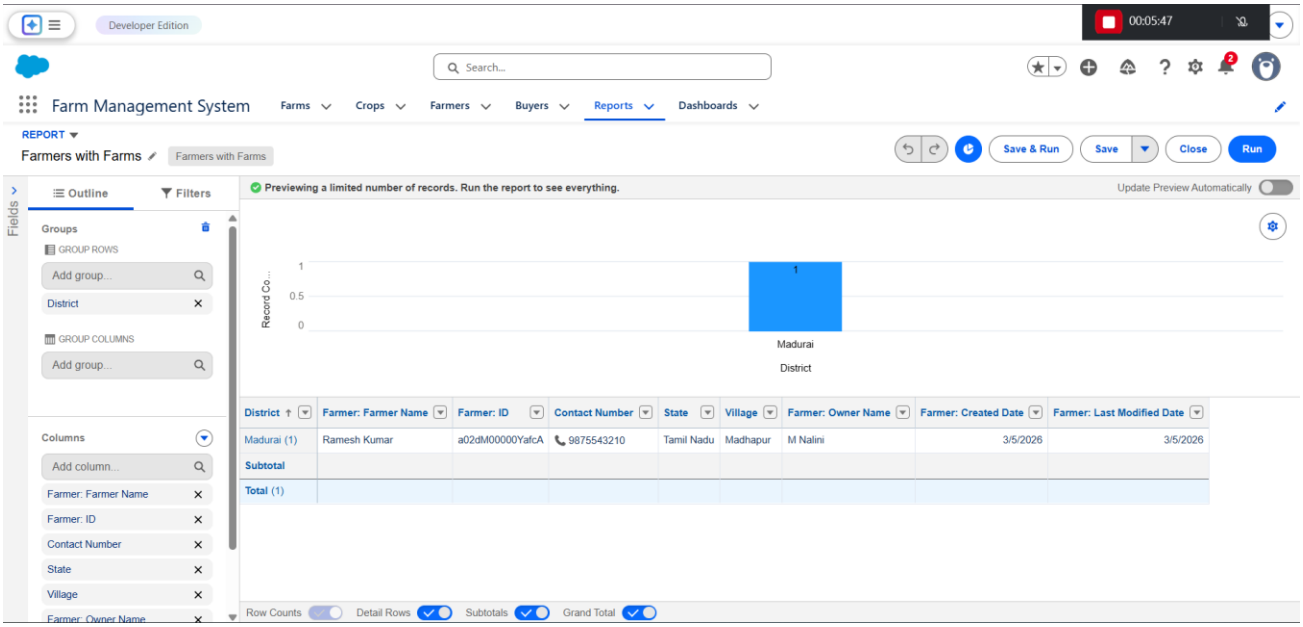


7. TESTING

During testing the following features were verified:

• Record creation
• Field validations
• Object relationships
• Profile and role access
• Sharing rules
• Report generation
• Dashboard display

Screenshots were captured to verify system functionality.


Developer Edition
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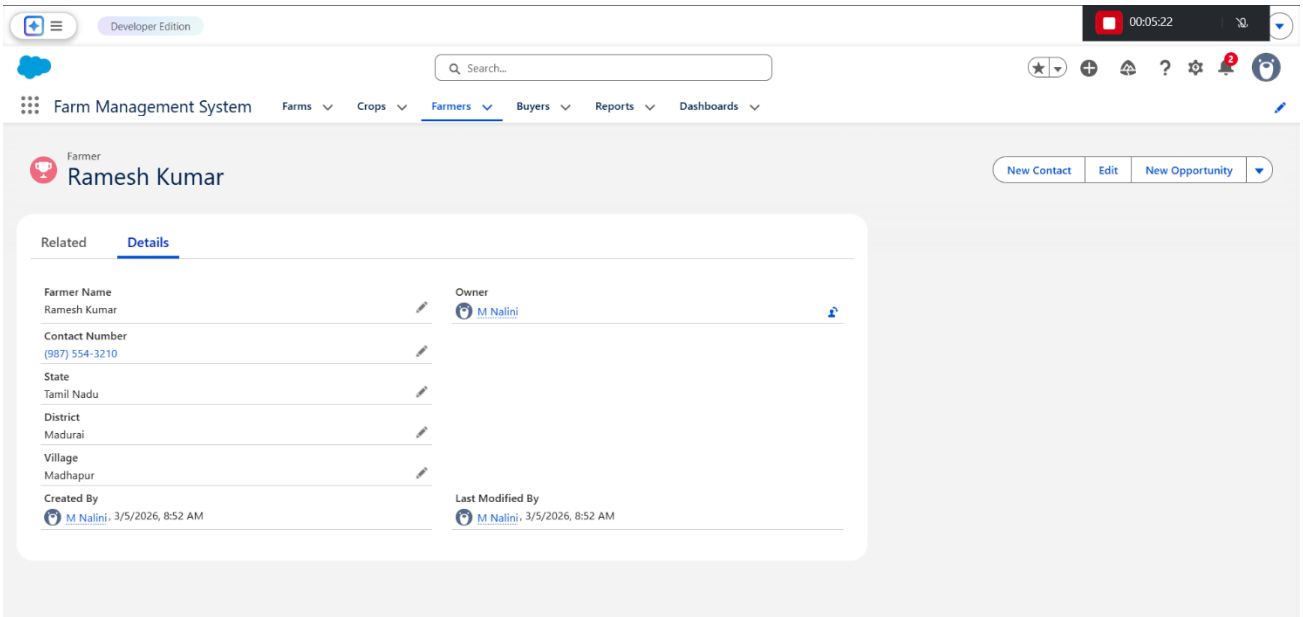
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
REPORT
Farmers with Farms
Farmers with Farms
Save & Run
Save
Close
Run
Fields
Outline
Filters
Previewing a limited number of records. Run the report to see everything.
Update Preview Automatically
Groups
GROUP ROWS
Add group...
District
GROUP COLUMNS
Add group...
Record Co...
1
0.5
0
1
Madurai
District
Columns
Add column...
Farmer: Farmer Name
Farmer: ID
Contact Number
State
Village
Farmer: Owner Name
District
Farmer: Farmer Name
Farmer: ID
Contact Number
State
Village
Farmer: Owner Name
Farmer: Created Date
Farmer: Last Modified Date
Madurai (1)
Ramesh Kumar
a02dM00000YafcA
9875543210
Tamil Nadu
Madhapur
M Nalini
3/5/2026
3/5/2026
Subtotal
Total (1)
Row Counts
Detail Rows
Subtotals
Grand Total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Farmers
Buyers
Reports
Dashboards
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New Contact
Edit
New Opportunity
Related
Details
Farmer Name
Ramesh Kumar
Owner
M Nalini
Contact Number
(987) 554-3210
State
Tamil Nadu
District
Madurai
Village
Madhapur
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Last Modified By
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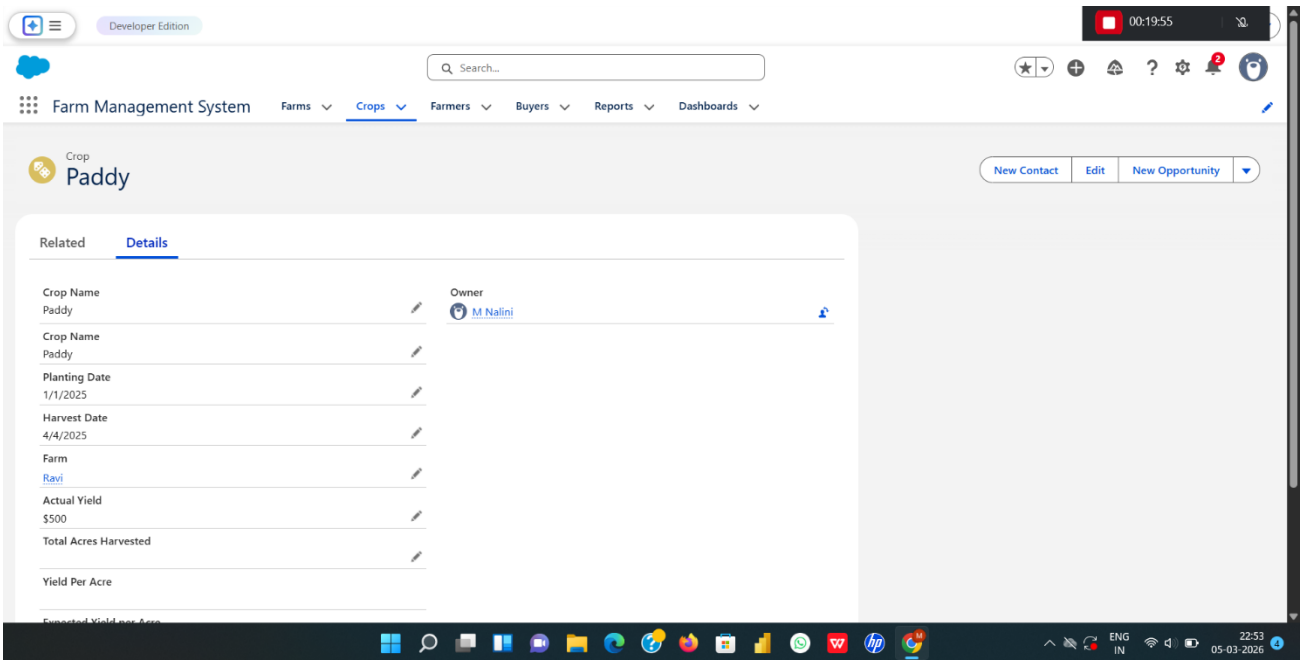
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Reports
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Related
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Paddy
Owner
M Nalini
Crop Name
Paddy
Planting Date
1/1/2025
Harvest Date
4/4/2025
Farm
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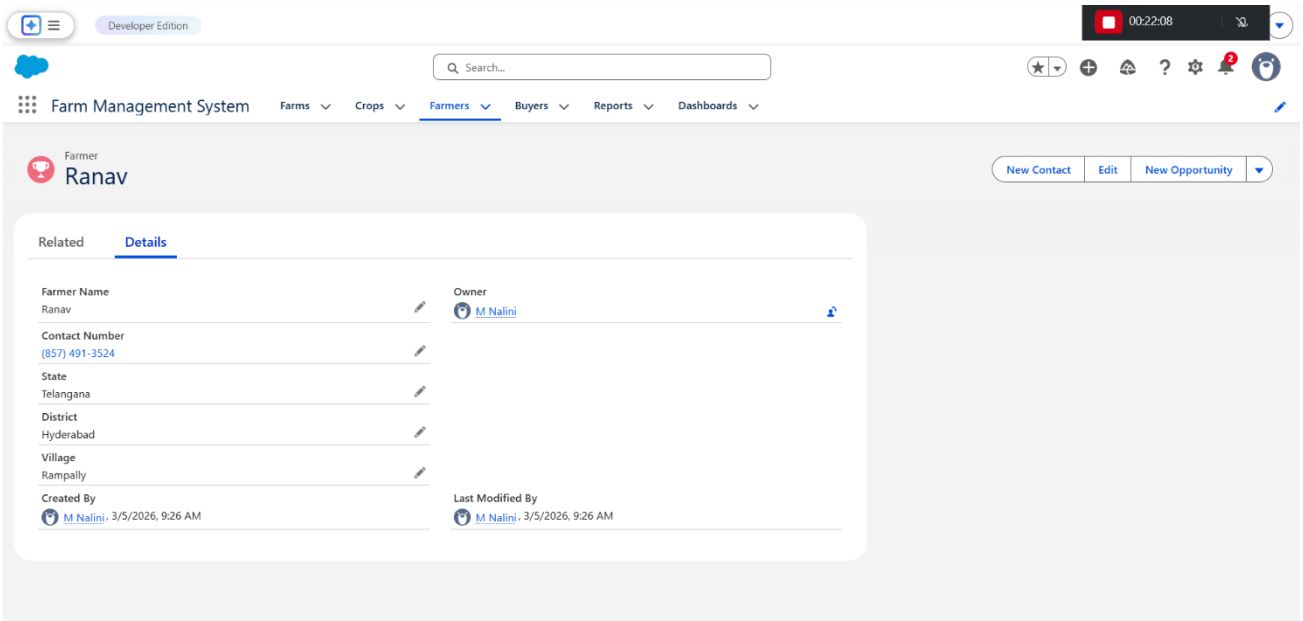
Total Acres Harvested
Yield Per Acre
ENG
IN
22:53
05-03-2026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Dashboards
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New Contact
Edit
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Related
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Farmer Name
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Owner
M Nalini
Contact Number
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State
Telangana
District
Hyderabad
Village
Rampally
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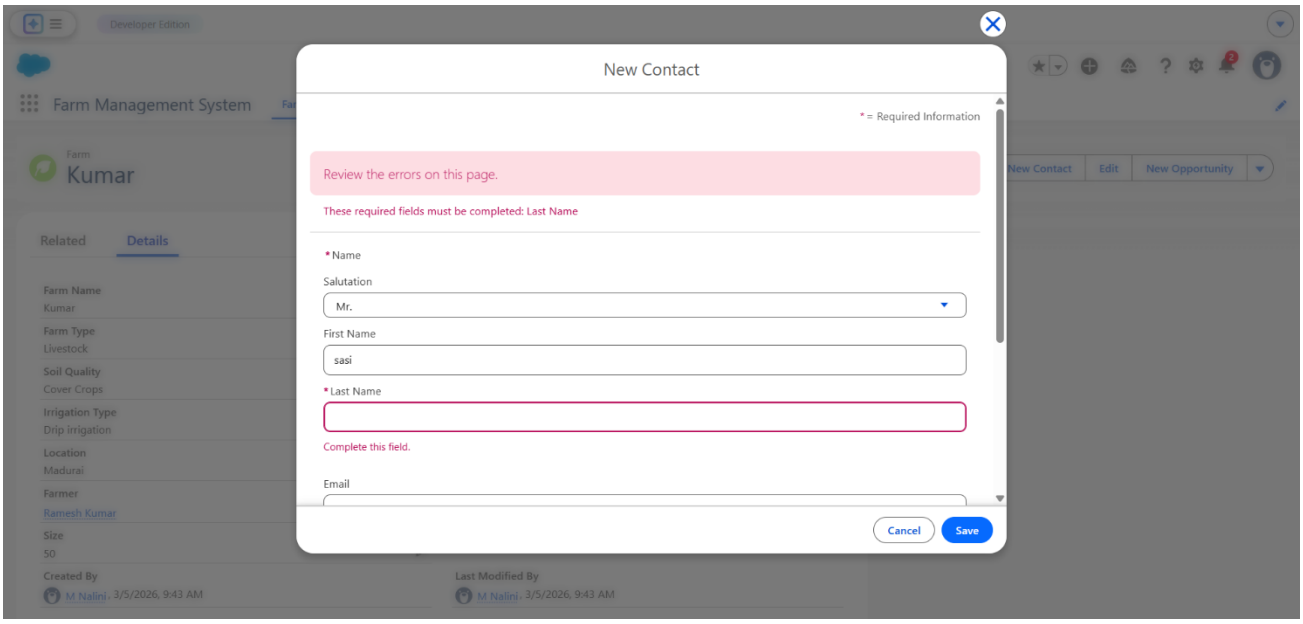

8. ADVANTAGES & DISADVANTAGES

Advantages

• Centralized data storage
• Reduced paperwork
• Improved accuracy
• Easy monitoring
• Real-time reporting

Disadvantages

• Requires internet connection
• Salesforce license limitations
• Limited offline functionality

9. CONCLUSION

The Farm Management System successfully digitizes agricultural data using Salesforce. It helps farmers and managers manage records efficiently, reduce errors, and analyze farm productivity through reports and dashboards.

10. FUTURE SCOPE

• Mobile application integration
• AI-based crop prediction
• Weather data integration
• IoT-based farm monitoring
• Market price analysis

11. APPENDIX

Tools Used

• Salesforce Developer Org
• Reports and Dashboards
• Custom Objects and Fields

Dataset

• Farm Records
• Crop Records
• Farmer Records
• Buyer Records